

Editorial

Learning to Appreciate Swiss Cheese and Other Industrial Engineering Concepts



We are pleased to serve as guest editors for this special “safety-focused” issue of *Practical Radiation Oncology*, and thank the American Society for Radiation Oncology (ASTRO) and Dr Lee for the opportunity. We received more than 30 submissions for this special issue, and all manuscripts were subject to the usual peer-review processes. This is a relatively large amount of content and reflects the vibrant manner in which our field has embraced quality and safety.

The evolving challenges related to safety have been recognized by many in our field,¹ but the issue of quality and safety gained great attention after a series of articles in *The New York Times* in 2010.²⁻⁵ This was followed by Congressional hearings, a public meeting with the Food and Drug Administration, an ASTRO/American Association of Physicists in Medicine (AAPM)-sponsored “Call to Action” meeting (filled to capacity), and ASTRO’s initiation of its Target Safety campaign. In parallel with many of these national activities, many individual institutions and groups expanded their own efforts in this area, and some of these efforts are highlighted in this special issue of *Practical Radiation Oncology*. Many of the submissions we received were acceptable for publication, but logistical matters prevented them all from being included in this special issue. Additional contributions will be seen in subsequent issues of the journal. Several concepts from industrial engineering can help us to put these safety initiatives in perspective.

The Swiss Cheese Model

In 1990, Dr James Reason proposed the Swiss Cheese Model to explain system failure.⁶ He suggested that although failure often appears to be due to an individual’s poor performance (eg, “human error”), performance is strongly influenced by the surrounding work environment and organizational priorities (Fig 1). Thus, understanding

the causes of errors requires one to consider—and minimizing the risk of errors requires one to address—each layer of the Swiss cheese. Focusing only on individual behaviors through incentives or discipline is a poor strategy. The reports in this special issue of *Practical Radiation Oncology* illustrate the importance of all layers of the Swiss cheese. It is important to emphasize that the different pieces of cheese are *not* intended to represent a series of sequential steps in a process (eg, it does *not* reflect consult → simulation/planning → treatment). Rather, as described as follows, the Swiss Cheese Model considers how organizational, environmental, and individual safeguards may be penetrated, resulting in an accident. Indeed, the model emphasizes that *suboptimally designed* layers of Swiss cheese may actually promote an accident.

- A. Organizational: Several reports note the importance of leadership in fostering continuous quality improvement initiatives—both by leader’s direct involvement and in helping to create an environment that enables initiatives to succeed. For example, Ford et al⁷ note the risks associated with the lack of clear organizational policies for critical procedures.
- B. Workplace: Several reports note the importance of having a physical workspace that supports continuous quality improvement initiatives. For example, the report from Cox et al⁸ notes the utility of an electronic board to track peer-defined recommended changes and teleconferencing to link their geographically separated centers. Similarly, Mazur et al⁹ note the importance of an easy-to-use Web-based tool to report and track reported incidents, and Kovalchuk et al¹⁰ note the utility of Visual Care Path within their electronic record to improve workflow efficiency and reduce errors.
- C. People: Several reports note the importance of broad participation of their faculty and staff in improvement work (eg, through broad attendance at their peer review sessions, receiving formal training in safety, in reporting incidents). The report from Shaffer et al highlights the need for people to have a heightened sense of safety mindfulness, especially as we apply new technologies in new ways.¹¹ They report a case with injury to the central biliary structures following stereotactic radiosurgery, despite the fact

Conflicts of interest: None.

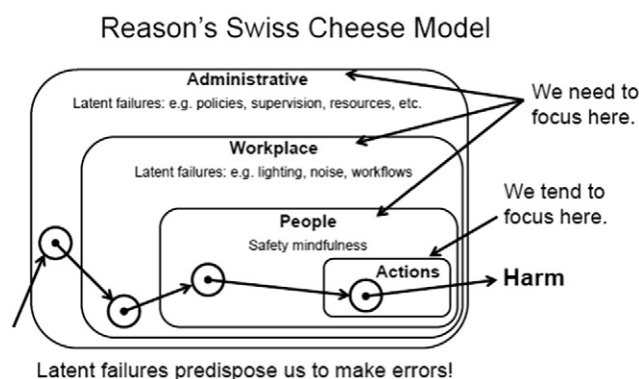


Figure 1 Venn-like diagram of Reason's Swiss Cheese Model. People do not take actions in a vacuum. Rather, people's actions are influenced by their own safety mindfulness and various environmental and organization factors as shown. Many safety initiatives focus only on people, but more attention should be paid to the "upstream" environmental and organizational factors. Additional outer layers reflecting (for example) societal, governmental, and professional factors could have also been drawn. Thus, actions taken at all of these "higher" levels impact people's performance and patient safety. Modified and adopted from Marks et al.²⁷

that customary "liver constraints" were well respected. This emphasizes that even architecturally parallel organs (eg, the liver) have critical components that behave in a serial manner and that the risk to these components may not be well predicted by our customary dose-volume constraints. A similar situation appears to exist in the lung, where the central hilar area is more sensitive to stereotactic body radiation therapy than is the (majority) peripheral lung.

Incident Learning

Several institutional reports in this issue of the journal (Nyflot et al,¹² Hill-Kayser et al,¹³ and Mazur et al⁹) focus on incident learning (ie, the reporting and tracking of events, concerns, performance issues, etc.), and highlight the utility of having objective quantitative data to help guide and assess improvement work. Several industries (eg, airlines, manufacturing) have successfully increased their reliability in part through incident learning; this actively involves workers in improving work, heightening their sense of safety mindfulness, and more accurately identifying potential safety problems (ie, the people at the front lines are more knowledgeable about what is actually going on). Quantification may be very helpful. For example, both Kovalchuk et al¹⁰ and Cox et al⁸ objectively quantify improvements in the timely completion of required tasks and in the quality of the work being done (eg, fewer errors detected, fewer changes needed) with their initiatives. Mazur et al⁹ quantify the apparent utility of safety barriers in each of the various steps of the treatment planning process. These sorts of objective quantitative data are most helpful helping to prioritize activities and to maintain adherence to (and enthusiasm for) continuing with ongoing processes.

Hierarchy of Effectiveness

Different types of initiatives to assure safety/quality have different rates of success.¹⁴ Typically, technology-based initiatives (eg, automation, forcing functions) are thought to be the most reliable, whereas standardizing human's work (eg, operating procedures, checklists, double-checks, reminder signs) is of intermediate utility, and policies and education are the least effective (Fig 2). Examples of interventions along this hierarchy in this issue include the report by Jani et al¹⁵ in which an automated system was used to verify patient identification and positioning and the report by Cox et al⁸ in which peer review, a form of "double check," was used to review contours and planning directives before initiation of treatment planning.

Normal Accident Theory

To effectively mitigate risks, one needs to understand how one's systems function and how they are likely to fail. The Normal Accident Theory is an attempt to do this and is widely accepted by scholars who study safety. This concept, as applied to errors associated with emergent radiation treatments, is well discussed by Chera et al¹⁶ in the accompanying commentary. Readers are encouraged to study that contribution because it explains this important concept.

Institutional Initiatives

The initiatives described at individual institutions are synergistic with several national initiatives. For example, ASTRO initiated its Target Safety plan in 2010.

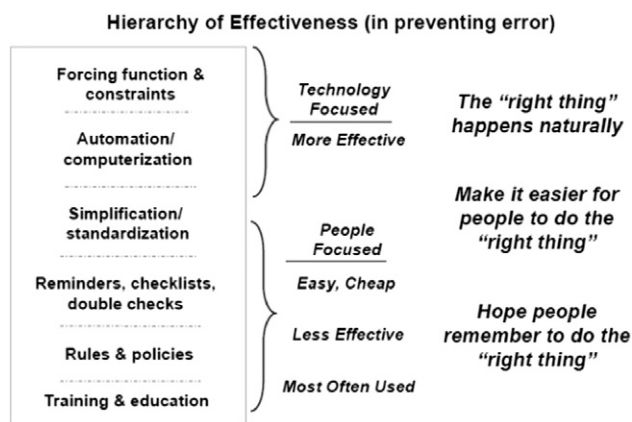


Figure 2 Hierarchy of effectiveness. Different approaches to modify behavior have different expectations for success, as shown. (Adapted from Dr Joseph Cafazzo, Healthcare Human Factors. Presentation at the Miami Safety Meeting, June 2010.) Reproduced with permission from Marks et al.²⁷

1. ASTRO/AAPM have successfully created a patient safety organization, the Radiation Oncology Incident Learning System (RO-ILS), to serve as a national repository for event reporting. Hoopes et al¹⁷ present the preliminary experience with RO-ILS in this issue. This program is administered through Clarity as a formal patient safety organization, and therefore all information submitted to RO-ILS has legal protections preventing discovery. Thus, RO-ILS will serve as a repository where we can pool experiences from centers around the country, learning from each other, without the fears often associated with the reporting of safety concerns. Industries (eg, aviation, nuclear power) and some medical fields (eg, anesthesiology) have successfully used this approach to promote reporting, learning from the pooled experience and hence guiding safety initiatives.

To date, 64 centers have joined RO-ILS (which equates to about 126 facilities) and 620 events have been reported (242 events that reached the patient, 206 that did not reach the patient, and 172 unsafe conditions). We, ASTRO, and AAPM encourage broad participation in RO-ILS. Providers/centers can participate in RO-ILS free of charge, and the legal structure of a patient safety organization affords protection from discovery by federal legislation. An old proverb is relevant in this setting: "Blessed is the one who learns from the mistakes of others since we do not live long enough to make every mistake ourselves."

2. ASTRO has launched its own practice accreditation program, Accreditation Program for Excellence (APEX), with standards that promote safety. For example, there are standards that encourage the use of checklists, standardized procedures, double checks, incident learning systems, safety education

and training, and broad participation in peer review with feedback. As practices seek accreditation through APEX, it is likely that their activities in all of these areas will increase, thereby improving safety.

3. ASTRO has published a series of white papers to provide guidance to our specialty to help promote safety in several high-risk areas (intensity modulated radiation therapy, image guided radiation therapy, stereotactic radiation surgery/stereotactic body radiation therapy, high-dose-rate radiation therapy) and peer review,¹⁸⁻²² with endorsement from various stakeholders including AAPM, American Association of Medical Dosimetrists, American Society of Radiologic Technologists, and American College of Radiology.
4. ASTRO and AAPM are working closely with our vendor colleagues to facilitate the safe transfer of data between various computer systems through the Integrating the Healthcare Enterprise-Radiation Oncology initiative. Beyond the clinical efficiencies of easy data transfer between systems, continued improvements in this area will improve safety.
5. ASTRO, AAPM, and other organizations have enhanced educational offerings addressing quality and safety at national meetings (eg, through keynote speakers, panels) and in their educational publications (eg, this special issue of *Practical Radiation Oncology* and the Safety is No Accident report). Innovative institutional educational programs have been developed as well.²³

Similarly, there are other initiatives both in the United States and internationally, for example, leveraging modern communications technology to provide peer-to-peer training and education for radiation therapy professionals (<https://i.treatsafely.org>) and peer review of clinical cases (<https://www.chartrounds.com/>). The Canadian

Partnership for Quality Radiotherapy promotes quality assurance/control guidelines, incident reporting, and patient engagement to improve safety. Additional similar international initiatives are sponsored by The Royal Australian and New Zealand College of Radiologists, the European Society for Radiation and Oncology, and the International Atomic Energy Agency.

Therefore, there is much to celebrate at an institutional, national, and international level. However, there is much more that we need to do. Indeed, many of the reports in this issue highlight the shortcomings of our current activities, and, therefore the need for change. For example, payers are increasingly focusing on “value” (commonly defined as quality divided by cost) and are asking specialties to define quality metrics. Metrics that directly reflect patient outcomes might be best. For example, thoracic surgeons report their postoperative mortality; clearly, this is a meaningful metric for patients. We and many other fields struggle in this regard. Most reports within radiation oncology typically do not use patient outcome–based metrics to assess quality. Rather, we often rely instead on so-called “process measures” (eg, minimizing the percent of ordered bone scans in patients with low-risk prostate cancer) or things such as patient satisfaction scores, staff safety culture survey results, or the number of reported incidents as evidence that we are “doing better.” Although measures of leadership,²⁴ culture,^{9,25–27} reporting,²⁸ and process²⁹ are likely related to quality and safety, this certainly is not always the case. Further, most process-based initiatives (eg, adding another checklist) can be detrimental because they take time, have opportunity costs, and add complexity that can increase the risk of error,³⁰ and thus should only be done if they add value. Accordingly, we need better measures that assess not only safety processes (eg, percent of cases undergoing peer review of image segmentation), but also safety-related outcomes (eg, percent of patients completing treatment as planned). More work needs to be done to identify safety interventions that result in clinically meaningful improvements in quality, safety, and associated metrics. Further, we need to define and aggressively better monitor process measures so that we can more fully understand *how* our own systems behave (eg, how workload relates to performance, how subsystems of our complex environment interact). Many industries have achieved “high reliability” (prevention of “major” accidents) by actively considering the many factors within all layers of the Swiss cheese. We cannot eliminate all human errors or all system failures, but we *can* eliminate serious patient harm.¹⁶ Our field needs to recognize, embrace, and address this challenge.

Our field enjoys a strong tradition of quality and safety. This is a tribute to our founding members who appreciated the power and risks of radiation and who instilled within the fabric of our field the need for careful oversight, attention to detail, and quality. As our field changes, we are facing increasing challenges in this area. Some of our

long-held traditional approaches to assuring quality may not be adequate, and we need to be creative in addressing these evolving challenges. The contributions in this special issue—reflecting initiatives at both the local and national levels—reflect our field’s strong commitment to continue to improve. We encourage everyone in the field to become more knowledgeable about the Swiss Cheese Model, the hierarchy of effectiveness, and other principles/terminology from industrial and safety engineering (eg, see [Appendix](#)). There is more applied research that needs to be done, much that we can learn, and much that we can do to adapt and apply these principles to further enhance safety; for ourselves, our patients, and society as a whole.

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Appendix

Many terms that we often used interchangeably have more specific meanings to experts in the field of quality and safety. A short glossary for several key terms is provided.²⁷ Using these terms in their more specific manner will help us understand the various factors that contribute to incidents and facilitate clear and unambiguous communication.

Defect: The lack of something in the system (ie, information, material, etc.) necessary or desirable for task completion.

Event (or Safety Event) is any of the following:

Incident that reached the patient: An incident that reaches the patient, with or without harm;

Near-miss: An incident that comes close to reaching the patient, but is caught and corrected beforehand; or

An unsafe condition: Any condition that increases the probability of an incident reaching the patient.

Hazard: Potential source of error that may lead to an event.

Harm: Physical injury or damage to the health of people, or damage to property or the environment.

Human error (or error): Slips, lapses, and mistakes are all considered forms of error.

Slip: Actively doing something unintended (this is observable); eg, inadvertently writing the prescription incorrectly.

Lapse: Failing to do something that was intended to be done (this is often *not* observable); eg, failing to write the prescription.

Mistakes: Purposeful action (perhaps done flawlessly) that was based on incorrect knowledge, or judgment; eg, prescribing radiation for a patient in a situation where it is not supported by the data.

Risk: Product of the “probability of occurrence of error” and “the severity of that error.”

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References

- Williamson JF, Thomadsen BR. Foreword. Symposium “Quality Assurance of Radiation Therapy: The Challenges of Advanced Technologies.” *Int J Radiat Oncol Biol Phys*. 2008;71(Suppl 1):S1.
- Bogdanich W. Safety features planned for radiation machines. Available at: <http://www.nytimes.com/2010/06/10/health/10radiation.html>. Accessed July 30, 2015.
- Bogdanich W. VA is fined over errors in radiation at hospital. Available at: <http://www.nytimes.com/2010/03/18/health/policy/18radiation.html>. Accessed July 30, 2015.
- Bogdanich W. Radiation offers new cures, and ways to do harm. Available at: <http://www.nytimes.com/2010/01/24/health/24radiation.html>. Accessed July 30, 2015.
- Bogdanich WRR. Radiation errors reported in Missouri. Available at: http://www.nytimes.com/2010/02/25/us/25radiation.html?_r=0. Accessed July 30, 2015.
- Reason J. Human error: Models and management. *BMJ*. 2000;320:768-770.
- Ford EC, Brown D, Donaldson H, et al. Patterns of practice for safety-critical processes in radiation oncology in the United States from the AAPM safety profile assessment survey. *Pract Radiat Oncol*. 2015;5:e423-e429.
- Cox BW, Kapur A, Sharma A, et al. Prospective contouring rounds: A novel, high-impact tool for optimizing quality assurance. *Pract Radiat Oncol*. 2015;5:e431-e436.
- Mazur L, Chera B, Mosaly P, et al. The association between event learning and continuous quality improvement programs and culture of patient safety. *Pract Radiat Oncol*. 2015;5:286-294.
- Kovalchuk N, Russo GA, Shin JY, Kachnic LA. Optimizing efficiency and safety in a radiation oncology department through the use of ARIA 11 Visual Care Path. *Pract Radiat Oncol*. 2015;5:295-303.
- Shaffer JL, Osmundon EC, Visser BC, Longacre TA, Koong AC, Chang DT. Stereotactic body radiation therapy and central liver toxicity: A case report. *Pract Radiat Oncol*. 2015;5:282-285.
- Nyflot MJ, Zeng J, Kusano AS, et al. Metrics of success: Measuring impact of a departmental near-miss incident learning system. *Pract Radiat Oncol*. 2015;5:e409-e416.
- Hill-Kayser CE, Gabriel P, Volz E, et al. Factors associated with event reporting in the pediatric radiation oncology population using an electronic incident reporting system. *Pract Radiat Oncol*. 2015;5:e417-e422.
- Leape LL. Error in medicine. *JAMA*. 1994;272:1851-1857.
- Jani SS, Low DA, Lamb JM. Automatic detection of patient identification and positioning errors in radiation therapy treatment using 3-dimensional setup images. *Pract Radiat Oncol*. 2015;5:304-311.
- Chera BS, Mazur L, Marks LB. Applying Normal Accident Theory to radiation oncology: Failures are normal but patient harm can be prevented. *Pract Radiat Oncol*. 2015;5:325-327.
- Hoopes DJ, Dicker AP, Eads NL, et al. RO-ILS: Radiation Oncology Incident Learning System: A report from the first year of experience. *Pract Radiat Oncol*. 2015;5:312-318.
- Thomadsen BR, Erickson BA, Eifel PJ, et al. A review of safety, quality management, and practice guidelines for high-dose-rate brachytherapy: Executive summary. *Pract Radiat Oncol*. 2014;4:65-70.
- Jaffray DA, Langen KM, Mageras G, et al. Safety considerations for IGRT: Executive summary. *Pract Radiat Oncol*. 2013;3:167-170.
- Marks LB, Adams RD, Pawlicki T, et al. Enhancing the role of case-oriented peer review to improve quality and safety in radiation oncology: Executive summary. *Pract Radiat Oncol*. 2013;3:149-156.
- Moran JM, Dempsey M, Eisbruch A, et al. Safety considerations for IMRT: Executive summary. *Med Phys*. 2011;38:5067-5072.
- Solberg TD, Balter JM, Benedict SH, et al. Quality and safety considerations in stereotactic radiosurgery and stereotactic body radiation therapy: Executive summary. *Pract Radiat Oncol*. 2012;2:2-9.
- Fogh S, Pawlicki T. A report on quality and safety education for radiation oncology residents. *Int J Radiat Oncol Biol Phys*. 2014;90:988-989.
- Clarke JR, Lerner JC, Marella W. The role for leaders of health care organizations in patient safety. *Am J Med Qual*. 2007;22:311-318.
- Mardon RE, Khanna K, Sorra J, Dyer N, Famolaro T. Exploring relationships between hospital patient safety culture and adverse events. *J Patient Saf*. 2010;6:226-232.
- Mann SM, Marchus R, Sachs B. Lessons from the cockpit: How team training can reduce errors on L&D. *Contemp OB/GYN*. 2006;51:34-40.
- Marks L, Mazur L, Chera B, Adams R. Engineering Patient Safety in Radiation Oncology: University of North Carolina’s Pursuit for High Reliability and Value Creation. Boca Raton, FL: CRC Press; 2015.
- Clark BG, Brown RJ, Ploquin J, Dunscombe P. Patient safety improvements in radiation treatment through 5 years of incident learning. *Pract Radiat Oncol*. 2013;3:157-163.
- International Atomic Energy Agency. Nuclear Power Plant Operating Experience From the IAEA/NEA International Reporting System for Operating Experience 2005–2008. Vienna, Austria: IAEA. 2010.
- Dekker S. Drift into Failure. Surrey, UK: Ashgate Publishing Limited; 2011.